



Physics Worksheet

Vectors

Name: _____ Student #: _____ Student Email #: _____

1. Algebra With Vectors

A boat is traveling across a river. It's going south east with a velocity of:

$\vec{v}_B = (-5, -4)\text{km h}^{-1}$. The river is flowing with a velocity of $\vec{v}_w = (2, 1)\text{km h}^{-1}$.

With what velocity is the boat crossing the river?

2. Scalar Product

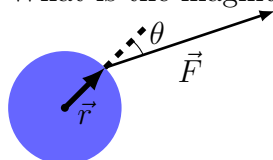
Aki and Luke go running off in different directions. $\vec{v}_A = (\frac{1}{3}, 2)\text{m s}^{-1}$,

$\vec{v}_L = (-1, \frac{3}{4})\text{m s}^{-1}$. What is the angle between their velocity vectors?

3. Cross Product

A force $\vec{F} = (6, 2)$ is applied to the edge of a disc centered at the origin at $\vec{r} = (1, 1)$.

What is the magnitude and direction of the angular acceleration?



4. Right Hand Rule

Use the right hand rule to find the direction $\vec{a} \times \vec{b}$ for the first two sets of vectors, and find the rotation vector of the third figure.

